THE COMPARISON MAP BETWEEN PULLBACK AND THE RIGHT ADJOINT IS COMPATIBLE WITH BASE CHANGE OF THE COMPARISON DIAGRAM

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file duality.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Preamble provenance: the pinned chapter file opens with an input of the shared preamble and carries no preamble text of its own, so the project's single Stacks-wide preamble staged at sources/stacks-duality/preamble.tex is excerpted verbatim under the recipe fixed by LANE_BRIEF 22.1, as three spans, lines 8-13 (verbatim package and the reference/slogan/history environments), lines 14-18 (Xy-pic setup) and lines 154-217 (the native theorem declarations and mathematical macros). Nothing was added, removed or redefined; the wrapper class and the amsmath/amssymb/amsthm/graphicx base packages are supplied by the assembler.

Editorial note. The counted statement and proof contain no cross reference, no citation and no included file, so the delivered body is self contained.

Editorial note. Canonical location of the counted proposition: Stacks tag 0B6P (sources/stacks-duality/pinned-original-source.tex line 2119).

Allowed standard prerequisites (editorial).

- The base change compatibility of the unit and of the trace map. Same chapter, Lemmas labelled lemma-unit-and-base-change and lemma-trace-map-and-base-change, cited at proof lines 2156-2162. Gives the commutativity for the first, third and fourth steps.
- The projection formula map and its compatibility with the base change diagram. Cohomology, Equation labelled cohomology-equation-projection-formula-map and Remark labelled cohomology-remark-compatible-with-diagram, cited at proof lines 2144-2148 and 2162-2166. Supplies the second step and its commutativity.
- The projection formula map is an isomorphism for the relevant morphisms. Derived Categories of Schemes, Lemma labelled perfect-lemma-cohomology-base-change, cited at proof lines 2146-2148. Makes the second step invertible.
- The comparison map and its description. Same chapter, equation labelled equation-compare-with-pullback and equation labelled equation-base-change-map, invoked in the statement. Fix the arrows whose compatibility is asserted.

Source reference. Original equation 0A9J: sources/stacks-duality/pinned-original-source.tex, source lines 927–927. Same-chapter pointer into the author environment equation at source line 927; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

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Source reference. Original equation 0A9S: sources/stacks-duality/pinned-original-source.tex, source lines 2079–2079. Same-chapter pointer into the author environment equation at source line 2079; invoked at proof line 2140 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 08HY: sources/stacks-chow/tags.txt, source lines 10796–10796. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0AA6: sources/stacks-duality/pinned-original-source.tex, source lines 1192–1192. Same-chapter pointer into the author environment equation at source line 1192; invoked at proof line None of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original equation 0B53: sources/stacks-chow/tags.txt, source lines 14021–14021. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2154 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 08EU: sources/stacks-chow/tags.txt, source lines 10685–10685. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2156 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6K: sources/stacks-duality/pinned-original-source.tex, source lines 1956–1956. Same-chapter pointer into the author environment lemma at source line 1956; invoked at proof line 2162 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original lemma 0B6J: sources/stacks-duality/pinned-original-source.tex, source lines 1867–1867. Same-chapter pointer into the author environment lemma at source line 1867; invoked at proof line 2163 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Source reference. Original reference 0B6B: sources/stacks-chow/tags.txt, source lines 14064–14064. Cross-chapter standard prerequisite located in the project tag list staged in the local snapshot; invoked at proof line 2166 of the counted proof. It is retained only as a navigation pointer; every load-bearing step is inside the delivered span.

Lemma 0.1. *Suppose we have a diagram (0A9J) where f and g are tor independent. Let $K \in D_{QCoh}(\mathcal{O}_Y)$. The diagram*

$$\begin{array}{ccc} L(g')^*(Lf^*K \otimes^{\mathbf{L}}_{\mathcal{O}_X} a(\mathcal{O}_Y)) & \longrightarrow & L(g')^*a(K) \\ \downarrow & & \downarrow \\ L(f')^*Lg^*K \otimes^{\mathbf{L}}_{\mathcal{O}_{X'}} a'(\mathcal{O}_{Y'}) & \longrightarrow & a'(Lg^*K) \end{array}$$

commutes where the horizontal arrows are the maps (0A9S) for K and Lg^*K and the vertical maps are constructed using Cohomology, Remark 08HY and (0AA6).

Proof. In this proof we will write f_* for Rf_* and f^* for Lf^*, etc, and we will write $\otimes$ for $\otimes^{\mathbf{L}}_{\mathcal{O}_X}$, etc. Let us write (0A9S) as the composition

$$\begin{aligned} f^*K \otimes a(\mathcal{O}_Y) &\to a(f_*(f^*K \otimes a(\mathcal{O}_Y))) \\ &\leftarrow a(K \otimes f_*a(\mathcal{O}_K)) \\ &\to a(K \otimes \mathcal{O}_Y) \\ &\to a(K) \end{aligned}$$

Here the first arrow is the unit η_f, the second arrow is a applied to Cohomology, Equation (0B53) which is an isomorphism by Derived Categories of Schemes, Lemma 08EU, the third arrow is a applied to $\mathrm{id}_K \otimes \mathrm{Tr}_f$, and the fourth arrow is a applied to the isomorphism $K \otimes \mathcal{O}_Y = K$. The proof of the lemma consists in showing that each of these maps gives rise to a commutative square as in the statement of the lemma. For η_f and Tr_f this is Lemmas 0B6K and 0B6J. For the arrow using Cohomology, Equation (0B53) this is Cohomology, Remark 0B6B. For the multiplication map it is clear. This finishes the proof. □